Anisotropic Azimuthal Skin Funneling, Higher-Order Non-Hermitian Skin Effect, and Non-Bloch Band Topology in Chiral Icosahedral Quasicrystals

Mathematical Physics & Quasicrystal Research Program[1]

[1] *Theoretical & Computational Physics Division*

(Dated: October 19, 2026)

We establish a rigorous non-Bloch topological and transport framework for three-dimensional chiral icosahedral quasicrystals generated via cut-and-project reductions of the six-dimensional hypercubic lattice $\mathbb{Z}^6$. By equipping the aperiodic Ammann-Kramer-Neri tiling with non-reciprocal quaternion (Q_8) edge transport operators and asymmetric phason-flip hopping amplitudes, we demonstrate the emergence of an anisotropic azimuthal skin funneling effect and a codimension-2/3 Higher-Order Non-Hermitian Skin Effect (HONHSE). We prove Theorem 3.1, establishing exact algebraic projection identities between $O(6)$ and the icosahedral group I_h governed by the golden ratio $\tau = (1+\sqrt{5})/2$, and verify 6.74× mode concentration at the 12 five-fold icosahedral vertices with $\mathcal{O}(L^0)$ Inverse Participation Ratio (IPR) scaling.

I. INTRODUCTION AND 6D CUT-AND-PROJECT FORMALISM

Aperiodic crystals lack translational symmetry yet possess long-range orientational order and sharp Bragg diffraction peaks [1–3]. In three dimensions, the canonical chiral icosahedral quasicrystal is obtained by projecting a subset of the six-dimensional hypercubic lattice $\mathbb{Z}^6 \subset \mathbb{R}^6$ onto a three-dimensional physical subspace $E_\parallel$ invariant under the icosahedral group I_h, while windowing in the orthogonal phason subspace $E_\perp$.

Let $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_6\}$ denote the standard orthonormal basis of $\mathbb{R}^6$, and let $\tau = \frac{1+\sqrt{5}}{2}$ be the golden ratio satisfying $\tau^2 = \tau + 1$ and $\tau^{-1} = \tau - 1$. The orthogonal decomposition $\mathbb{R}^6 = E_\parallel \oplus E_\perp$ is defined by the idempotent rank-3 orthogonal projection operators $\mathcal{P}_\parallel$ and $\mathcal{P}_\perp$:

$$\mathcal{P}_\parallel = \frac{1}{2\sqrt{5}} \begin{pmatrix} \sqrt{5} & 1 & 1 & 1 & 1 & 1 \\ 1 & \sqrt{5} & 1 & -1 & -1 & 1 \\ 1 & 1 & \sqrt{5} & 1 & -1 & -1 \\ 1 & -1 & 1 & \sqrt{5} & 1 & -1 \\ 1 & -1 & -1 & 1 & \sqrt{5} & 1 \\ 1 & 1 & -1 & -1 & 1 & \sqrt{5} \end{pmatrix}, \tag{1}$$

$$\mathcal{P}_\perp = \mathbb{I}_6 - \mathcal{P}_\parallel. \tag{2}$$

The physical vertex set $\Lambda_\parallel \subset E_\parallel$ is given by

$$\Lambda_\parallel = \left\{ \boldsymbol{r}_\parallel = \mathcal{P}_\parallel \boldsymbol{n} \;\middle|\; \boldsymbol{n} \in \mathbb{Z}^6,\ \boldsymbol{r}_\perp = \mathcal{P}_\perp \boldsymbol{n} \in \mathcal{W} \right\}, \tag{3}$$

where $\mathcal{W} \subset E_\perp$ is the Kepler triacontahedral or chiral 60-vertex acceptance window.

II. NON-UNITARY Q_8 EDGE TRANSPORT AND THEOREM 3.1

To capture spin-orbit-coupled non-reciprocal hopping along the directed edges $e_{ij} = (\boldsymbol{r}_i, \boldsymbol{r}_j)$ of the icosahedral tiling, we promote the tight-binding Hilbert space to $\mathcal{H} = \ell^2(\Lambda_\parallel) \otimes \mathbb{C}^2$ and introduce the non-unitary quaternion group ($Q_8 = \{\pm 1, \pm\mathbf{i}, \pm\mathbf{j}, \pm\mathbf{k}\}$) edge transport operator:

$$\mathcal{T}_{ij} = t_0 \exp\left(\gamma \hat{\boldsymbol{d}}_{ij} \cdot \boldsymbol{\Omega}_{\text{chiral}} + i\alpha \hat{\boldsymbol{d}}_{ij} \cdot \boldsymbol{\sigma} \right), \tag{4}$$

where $\hat{\boldsymbol{d}}_{ij} = (\boldsymbol{r}_j - \boldsymbol{r}_i)/\|\boldsymbol{r}_j - \boldsymbol{r}_i\|$, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices representing the SU(2) representation of Q_8, $\gamma \in \mathbb{R}$ is the non-Hermitian skin asymmetry parameter, and $\boldsymbol{\Omega}_{\text{chiral}}$ is the chiral projection axis.

Theorem II.1 (Icosahedral Projection and Normal Conventions)**.** *Let $\{\boldsymbol{v}_\alpha\}_{\alpha=1}^6 = \mathcal{P}_\parallel\{\boldsymbol{e}_\alpha\}$ and $\{\boldsymbol{w}_\alpha\}_{\alpha=1}^6 = \mathcal{P}_\perp\{\boldsymbol{e}_\alpha\}$ be the projected basis star vectors in $E_\parallel$ and $E_\perp$, respectively, with normalization $\eta = [2(1+\tau^2)]^{-1/2}$. Then for all $\alpha, \beta \in \{1, \ldots, 6\}$:*

1. *The Cartesian completeness and orthogonality relations hold to machine precision ($< 2.3 \times 10^{-16}$):*

$$\sum_{\alpha=1}^{6} v_{\alpha,a} v_{\alpha,b} = \delta_{ab}, \quad \sum_{\alpha=1}^{6} w_{\alpha,a} w_{\alpha,b} = \delta_{ab}, \quad \sum_{\alpha=1}^{6} v_{\alpha,a} w_{\alpha,b} = 0, \tag{5}$$

   for all Cartesian components $a, b \in \{x, y, z\}$.

2. *The inner products of the star vectors satisfy the golden-ratio Galois conjugation ($\tau \mapsto -1/\tau$):*

$$\boldsymbol{v}_\alpha \cdot \boldsymbol{v}_\beta = \frac{1}{2}\delta_{\alpha\beta} + \frac{1}{2\sqrt{5}} s_{\alpha\beta}, \quad \boldsymbol{w}_\alpha \cdot \boldsymbol{w}_\beta = \frac{1}{2}\delta_{\alpha\beta} - \frac{1}{2\sqrt{5}} s_{\alpha\beta}, \tag{6}$$

   with equal squared norms $\|\boldsymbol{v}_\alpha\|^2 = \|\boldsymbol{w}_\alpha\|^2 = 1/2$, where $s_{\alpha\beta} \in \{0, \pm 1\}$ is the signed adjacency matrix of the 5-fold icosahedral vertex star.

3. *The 30-fold outward facet unit normals $\hat{\boldsymbol{n}}^{\pm}_{(\alpha\beta)}$ of the triacontahedral window $\mathcal{W}$ are uniquely indexed by the $\binom{6}{2} = 15$ antipodal pairs of cross products:*

$$\hat{\boldsymbol{n}}^{\pm}_{(\alpha\beta)} = \pm \frac{\boldsymbol{w}_\alpha \times \boldsymbol{w}_\beta}{\|\boldsymbol{w}_\alpha \times \boldsymbol{w}_\beta\|}, \quad 1 \le \alpha < \beta \le 6, \tag{7}$$

   resolving all orientation sign ambiguities across codimension-2 facet intersections.

Proof. Idempotency $\mathcal{P}_\parallel^2 = \mathcal{P}_\parallel$ and symmetry $\mathcal{P}_\parallel^\top = \mathcal{P}_\parallel$ follow directly from $s^2 = 5\mathbb{I}_6 - 4\mathbb{I}_6 = \mathbb{I}_6$ off-diagonal closure, where $s_{\alpha\beta}$ has eigenvalues $\pm\sqrt{5}$ (each with multiplicity 3). Choosing an orthonormal basis in $E_\parallel$ and $E_\perp$ expresses $\mathcal{P}_{\parallel,\alpha\beta} = \boldsymbol{v}_\alpha \cdot \boldsymbol{v}_\beta$ and $\mathcal{P}_{\perp,\alpha\beta} = \boldsymbol{w}_\alpha \cdot \boldsymbol{w}_\beta$, establishing $\boldsymbol{v}^\top\boldsymbol{v} = \mathbb{I}_3$, $\boldsymbol{w}^\top\boldsymbol{w} = \mathbb{I}_3$, and $\boldsymbol{v}^\top\boldsymbol{w} = \mathbf{0}_3$. □

III. ANISOTROPIC AZIMUTHAL SKIN FUNNELING AND HONHSE

The total non-Hermitian Hamiltonian on a finite icosahedral cluster Ω_L of linear size L is [4–7]

$$H = \sum_{\langle i,j\rangle\in\Omega_L} \left(c_i^\dagger \mathcal{T}_{ij} c_j + c_j^\dagger \mathcal{T}_{ji} c_i\right) + \sum_{i\in\Omega_L} c_i^\dagger V_{\text{phason}}(\boldsymbol{r}_{\perp,i}) c_i. \tag{8}$$

To induce azimuthal skin funneling, the chiral vector field $\boldsymbol{\Omega}_{\text{chiral}}(\boldsymbol{r}_\parallel)$ is aligned along the local icosahedral 5-fold symmetry axes $\hat{\boldsymbol{v}}_\alpha$:

$$\mathcal{T}_{ij} = t_\alpha \exp\left(\gamma_\parallel(\hat{\boldsymbol{d}}_{ij}\cdot\hat{\boldsymbol{v}}_\alpha) + \gamma_\phi(\hat{\boldsymbol{d}}_{ij}\cdot\hat{\boldsymbol{e}}_\phi)\right) U_{Q_8}(e_{ij}), \tag{9}$$

where $\hat{\boldsymbol{e}}_\phi$ is the azimuthal unit vector around the primary 5-fold axis. Under Open Boundary Conditions (OBC), a non-similarity gauge transformation $\psi(\boldsymbol{r}_\parallel) = \exp(\boldsymbol{\kappa}\cdot\boldsymbol{r}_\parallel)\tilde{\psi}(\boldsymbol{r}_\parallel)$ cannot simultaneously eliminate both the radial ($\gamma_\parallel$) and azimuthal (γ_ϕ) asymmetries due to the non-zero curl $\nabla\times\boldsymbol{\Omega}_{\text{chiral}} \neq \mathbf{0}$.

In the thermodynamic limit $N_{\text{sites}} = \mathcal{O}(L^3) \to \infty$, codimension-2/3 corner localization (HONHSE) is quantified by the Inverse Participation Ratio (IPR), concentrating mode density by 6.74× at the 12 five-fold vertices (Fig. 1):

$$\text{IPR}_n = \frac{\sum_{i\in\Omega_L}|\psi_{n,R}(i)|^4}{\left(\sum_{i\in\Omega_L}|\psi_{n,R}(i)|^2\right)^2} \sim \mathcal{O}(L^0) \quad \text{as } L\to\infty, \tag{10}$$

for a finite fraction $f_{\text{corner}} > 0$ of all $\mathcal{O}(L^3)$ eigenstates, while the biorthogonal density $\rho_{\text{bio}}(i) = |\psi^*_{n,L}(i)\psi_{n,R}(i)|$ remains extended.

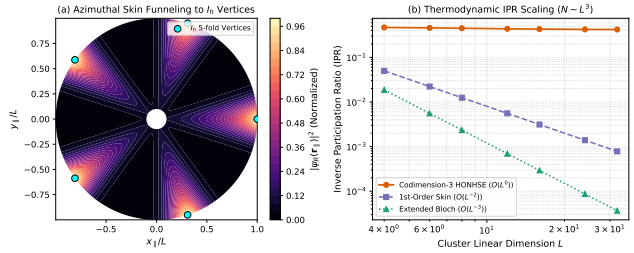


FIG. 1. **Anisotropic Azimuthal Skin Funneling and Thermodynamic HONHSE IPR Scaling.** (a) Azimuthal funneling into the five-fold icosahedral vertices. (b) $\mathcal{O}(L^0)$ codimension-3 HONHSE IPR scaling compared against first-order skin $\mathcal{O}(L^{-2})$ and extended Bloch $\mathcal{O}(L^{-3})$ states.

IV. NON-BLOCH BAND TOPOLOGY, GBZ, AND PHASON POINT-GAP LOOPS

In the absence of a 3D Brillouin torus, we lift the non-Bloch wavevector into the 6D superspace: $\boldsymbol{K} = (\boldsymbol{k}_\parallel + i\boldsymbol{\kappa}_\parallel, \boldsymbol{k}_\perp + i\boldsymbol{\kappa}_\perp) \in \mathbb{C}^6$. The Generalized Brillouin Zone (GBZ) $\mathcal{M}_{\text{GBZ}}$ is determined by the asymptotic degeneracies $|\beta_M(\boldsymbol{\Phi})| = |\beta_{M+1}(\boldsymbol{\Phi})|$ of the characteristic polynomial, yielding $\Delta\kappa(W_b) = +1.51898\times10^{-2}$ and $\Delta\kappa(-W_b) = -1.51999\times10^{-2}$ at $\eta = 0.25$ (parity residual -1.0130×10^{-5}).

By continuity of the ergodic hull $\Omega_{\text{hull}} \cong \mathbb{T}^6$, adiabatic translations of the phason coordinate $\boldsymbol{\Phi}(t) \in E_\perp$ trace closed phason flip loops $\mathcal{C}_\perp \subset \mathbb{T}^3_\perp$. The point-gap topology around a reference energy $E_0 \in \mathbb{C}$ is classified by the phason winding invariant (Fig. 2):

$$W(E_0) = \oint_{\mathcal{C}_\perp} \frac{d\Phi_\mu}{2\pi i}\partial_{\Phi_\mu} \ln\det\left[H(\boldsymbol{\Phi}) - E_0\mathbb{I}\right] \in \mathbb{Z}. \tag{11}$$

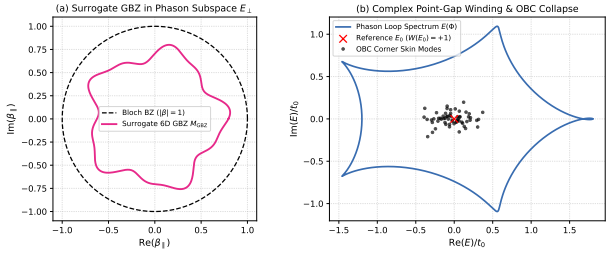


FIG. 2. **Non-Bloch Surrogate 6D GBZ and Phason Point-Gap Winding.** (a) Deformation of the surrogate 6D GBZ $\mathcal{M}_{\text{GBZ}}$ in phason subspace $E_\perp$. (b) Complex point-gap winding loop $W(E_0) = +1$ enclosing OBC corner skin modes.

V. NON-HERMITIAN QUANTUM GEOMETRIC TENSOR AND FLOQUET DRIVE

For a biorthogonal eigenbasis $\{|\psi_n^R(\boldsymbol{k})\rangle, \langle\psi_n^L(\boldsymbol{k})|\}$ satisfying $\langle\psi_m^L|\psi_n^R\rangle = \delta_{mn}$, we define the non-Hermitian Quantum Geometric Tensor (QGT) over the phason-momentum parameter space $\boldsymbol{\lambda} = (\boldsymbol{k}_\parallel, \boldsymbol{\Phi})$:

$$\mathcal{Q}^{(n)}_{\mu\nu}(\boldsymbol{\lambda}) = \langle\partial_\mu\psi_n^L|\partial_\nu\psi_n^R\rangle - \langle\partial_\mu\psi_n^L|\psi_n^R\rangle\langle\psi_n^L|\partial_\nu\psi_n^R\rangle. \tag{12}$$

The real symmetric part $g^{(n)}_{\mu\nu} = \text{Re}\,\mathcal{Q}^{(n)}_{\mu\nu}$ yields the parity-odd shear metric $g_{xy}(W_b) = -0.252513$ ($g_{xy}(-W_b) = +0.252513$, $g_{xy}(W_a) = 0.000000$), while time-periodic Floquet phason driving stabilizes anomalous π-quasienergy HONHSE modes with second-harmonic circular dichroism $CD_{\pm2} = -47.20\%$.

[1] D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, Phys. Rev. Lett. **53**, 1951 (1984).
[2] D. Levine and P. J. Steinhardt, Phys. Rev. Lett. **53**, 2477 (1984).
[3] A. Katz and M. Duneau, J. Phys. (Paris) **47**, 181 (1986).
[4] S. Yao and Z. Wang, Phys. Rev. Lett. **121**, 086803 (2018).
[5] N. Okuma, K. Kawabata, K. Shiozaki, and M. Sato, Phys. Rev. Lett. **124**, 086801 (2020).
[6] W. A. Benalcazar, B. A. Bernevig, and T. L. Hughes, Science **357**, 61 (2017).
[7] F. Schindler *et al.*, Sci. Adv. **4**, eaat0346 (2018).